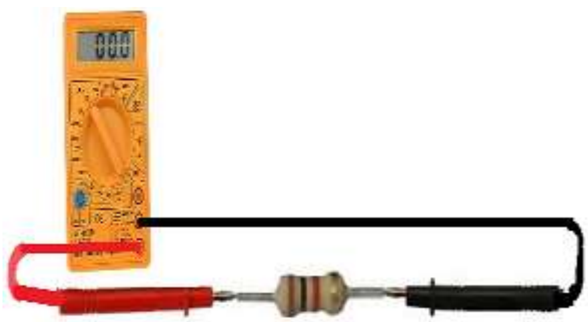
HOW TO CHECK ELECTRONIC COMPONENTS

How to Test a Resistor?

To check to see whether a resistor is good or not, we need to only perform one test and this is to check the resistor's resistance value, using the ohmmeter of a multimeter.

Test a Resistor with an Ohmmeter:

Testing a Resistor with an ohmmeter is the best, easiest and most effective way to tell whether a resistor is good or not.To set up for the check, we take the ohmmeter and place its probes across the leads of the resistor. The orientation doesn't matter, because resistance isn't polarized.



The resistance that the ohmmeter reads should be close to the rated resistance of the resistor.
If the ohmmeter is reading in the value and tolerance range of the resistor, the resistor is good. If the ohmmeter is reading (especially drastically) outside of this range, the resistor is defective and should be replaced.

How to Test whether a Resistor is Open:

If a resistor is reading a very high resistance, above its rated value, it is open. It is defective and, thus, should be replaced.

How to Test whether a Resistor is Shorted:

If a resistor is reading a very low resistance, near 0Ω, it's shorted internally. It is defective and, thus, should be replaced.

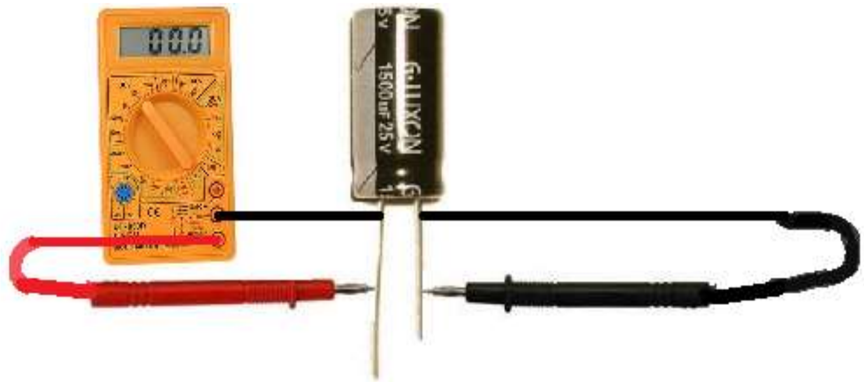
A resistance test is the only test that is needed to determine whether a resistor is good. If you want to examine more advanced features of a resistor, then additional tests may be necessary, but for all basic purposes, this test is sufficient for checking resistors.

How to Test a Capacitor?

There are many checks we can do to see if a capacitor is functioning the way it should.

Test a Capacitor with an Ohmmeter of a Multimeter:

A very good test you can do is to check a capacitor with your multimeter set on the ohmmeter setting. By taking the capacitor's resistance, we can determine whether the capacitor is good or bad. To do this test, we take the ohmmeter and place the probes across the leads of the capacitor. The orientation doesn't matter.



If we read a very low resistance (near 0Ω) across the capacitor, we know the capacitor is defective. It is reading as if there is a short circuit across it.

If we read a very high resistance across the capacitor (several MΩ), this is a sign that the capacitor likely is defective as well. It is reading as if there is an open circuit across the capacitor.

A normal capacitor would have a resistance reading up somewhere in between these 2 extremes, say, anywhere in the tens of thousands or hundreds of thousands of ohms. But not 0Ω or several MΩ. This is a simple but effective method for finding out if a capacitor is defective or not.

Test a Capacitor with a Multimeter in the Capacitance Setting:

Another check you can do is check the capacitance of the capacitor with a multimeter, if you have a capacitance meter on your multimeter. All you have to do is read the capacitance that is on the exterior of the capacitor and take the multimeter probes and place them on the leads of the capacitor. Polarity doesn't matter. This is the same as the how the setup is for the first illustration, only now the multimeter is set to the capacitance setting.

You should read a value near the capacitance rating of the capacitor. Due to tolerance and the fact that (specifically, electrolytic capacitors) may dry up, you may read a little less in value than the capacitance of the rating. This is fine. If it is a little lower, it is still a good capacitor. However, if you read a significantly lower capacitance or none at all, this is a sure sign that the capacitor is defective and needs to be replaced.

Checking the capacitance of a capacitor is a great test for determining whether a capacitor is good or not.

Test a Capacitor with a Voltmeter

Another test you can do to check if a capacitor is good or not is a voltage test.

After all, capacitors are storage devices. They store a potential difference of charges across their plate, which are voltages.

A test that you can do is to see if a capacitor is working as normal is to charge it up with a voltage and then read the voltage across the terminals. If it reads the voltage that you charged it to, then the capacitor is doing its job and can retain voltage across its terminals. If it is not charging up and reading voltage, this is a sign the capacitor is defective.

To charge the capacitor with voltage, apply DC voltage to the capacitor leads. Now polarity is very important for polarized capacitors (electrolytic capacitors). If you are dealing with a polarized capacitor, then you must observe polarity and the correct lead assignments. Positive voltage goes to the anode (the longer lead) of the capacitor and negative or ground goes to the cathode (the shorter lead) of the capacitor. Apply a voltage which is less than the voltage rating of the capacitor for a few seconds.

For example, feed a 25V capacitor 9 volts and let the 9 volts charge it up for a few seconds. As long as you're not using a huge, huge capacitor, then it will charge in a very short period of time, just a few seconds. After the charge is finished, disconnect the capacitor from the voltage source and read its voltage with the multimeter.

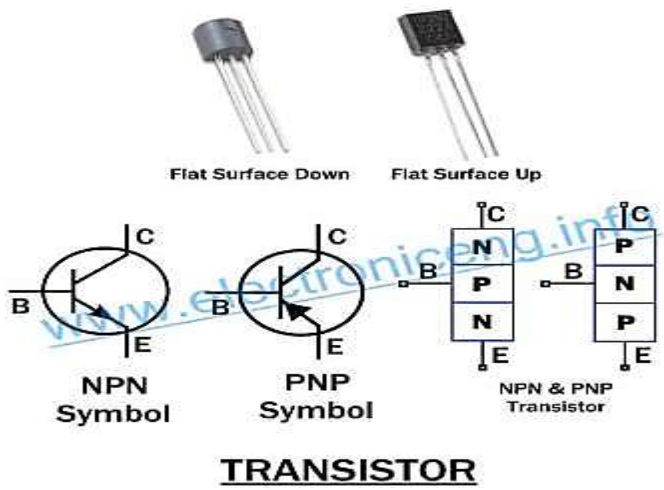
The voltage at first should read near the 9 volts (or whatever voltage) you fed it. Note that the voltage will discharge rapidly and head down to 0V because the capacitor is discharging its voltage through the multimeter. However, you should read the charged voltage value at first before it rapidly declines.

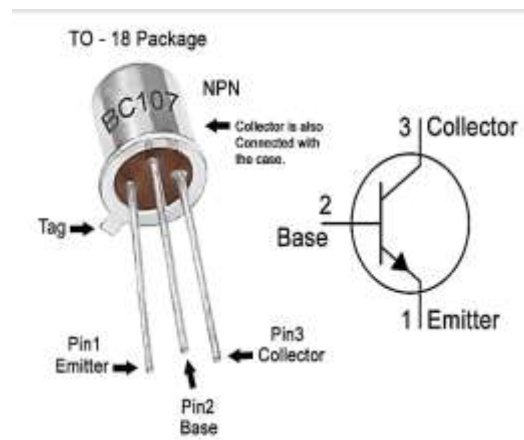
TRANSISTOR TEST FOR TERMINAL IDENTIFICATION, TYPE & CONDITION

This test is only applicable for **BJT** transistors. So before any transistor test, we need to know about the **structure** of the **BJT.** This transistor test helps you in the **terminal's identification, NPN, PNP &Good/Damaged transistor**

Transistor (BJT)

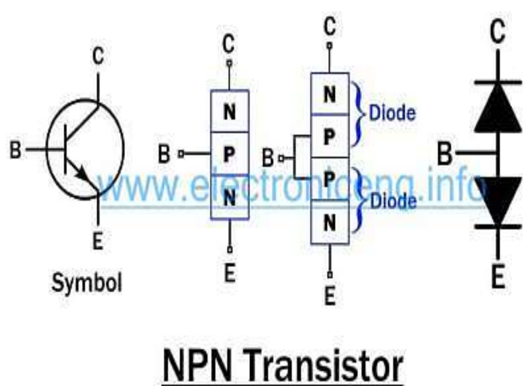
A BJT (Bipolar Junction Transistor) is a three-terminal semiconductor device. It's made up of two **P-N** junction diodes fused together forming three layers known as **Base, Emitter & Collector**.





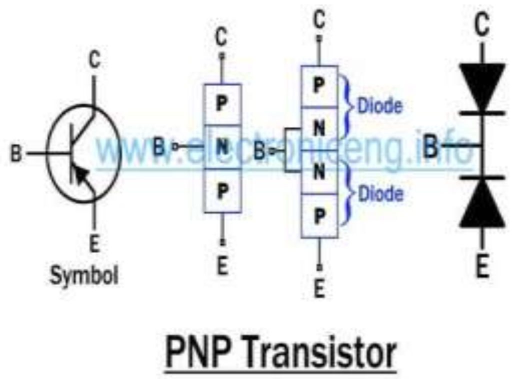
There are two types of transistors base on the polarity of its layers.

NPN :



In this BJT, the **Base** i.e. the **P-doped** layer is sandwiched between **N-doped** layers known as **Collector & Emitter**. The difference between the Collector & Emitter is that the Emitter is the **heavily doped** layer.
NPN corresponds to two diodes being fused together by the Anode terminal as shown in the figure below.

PNP:



PNP transistor is made up of an **N-doped** layer (**Base**) sandwiched between **P-doped** layers known as **Collector & Emitter**.PNP transistor corresponds to two diodes, the cathode terminal of these two diodes is fused together as shown in the figure below.

This transistor test uses the Diode test function of the Multimeter. So, For the sake of this transistor test, you need to know about the diode test**.**

Diode test mode:

Forward bias P-N junction: Multimeter reads some voltage & beeps.
Reverse bias P-N junction: Multimeter reads OL (Over Limit)

Terminal's Identification

The first step in the transistor test is to identify the terminals (**Base, Emitter & Collector**) of the transistor.

First, you need to mark the terminals of the transistor with numbers **1,2,3**. In order to do that, hold the transistor's flat side facing towards you and start from the left side as shown in the figure below.

Base Terminal Identification

- Put Multimeter in **diode test mode**.
- Place black (common) probe & red probe on any two terminals at a time.
- Test all possible terminal combinations i.e. **1-2**, **1-3**, **2-1**, **2-3**, **3-1**, **3-2**.
- Two of these combinations should pass diode test (reading shows voltage **0.5v to 0.8v**), the **common terminal** in these two combinations is the **Base terminal**.

Suppose, **2-1** & **2-3** combinations pass diode test then **2** is the base terminal

Emitter & Collector Identification

With the successful identification of the base terminal, two terminals (**1 & 3**) remain unknown. if you identify the second terminal, subsequently you will also know the third terminal.

- Set the Multimeter in **diode test mode**.
- Record the voltage reading of base terminal with both terminals **1 & 3** one by one.
  - terminal having a **higher voltage** between the two is the **Emitter**.
  - The terminal with **lower voltage** compared to the other is **Collector**.

In this example, suppose the 2-1 voltage reading = 0.6v & 2-3 voltage reading = 0.7v
So the Emitter is terminal 3 & Collector is terminal 1.

Type: NPN or PNP

The next step in the **transistor test** is to know whether the transmitter is **NPN** or **PNP**.
This step depends on the results of the above transistor test.

NPN Test

- Put Multimeter in **diode test mode**.
- Place the **Red** (positive) probe on **Base terminal** & the **black** (common or negative) terminal on **Emitter** & **Collector** one by one.
- If they pass the diode test, it means the junctions are forward bias & it is an **NPN** transistor.

If you don't know the terminals.

- Set the multimeter in diode test mode.
- Test all the six combinations of terminals for the diode test.
- Note those two combinations, whose diode test is positive (multimeter Beeps or show voltage). If the common terminal in these two combinations is connected to the **red probe** of the multimeter, the transistor is **NPN**.

PNP Test

PNP transistor test is a little different than the **NPN** transistor test.

- Put the Multimeter in **diode test mode**.
- Connect the **Black** (common) probe with **Base** & the **Red** probe with **Emitter** & **Collector** one at a time.
- If these both combinations pass the diode test, the Transistor is **PNP**.

If you don't know the terminals.

- Check all (six) possible combinations of the terminals for the **diode test**.

- Note the **two combinations** that pass the diode test.
- If the **common terminal** in these two combinations is connected to the **Black** or common probe of the Multimeter, the transistor is **PNP**.

<u>Testing The Transistor (Good or Damaged)</u>

This transistor test helps us in identifying if the transistor is **good or damaged**.
Set the Multimeter in **diode test mode** and test all the possible combination for the diode test.
Note down the reading for each combination.
If the transistor satisfies the readings given in the chart below, it is **good**. If the readings don't match with the chart below, the transistor is damaged & need to be **replaced**

DIODE TEST MODE

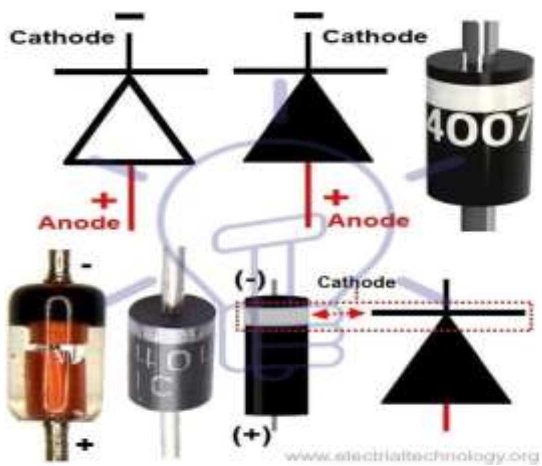
Multimeter's Probes		Multimeter's Reading	
Red Probe	Black Probe	For NPN	For PNP
Base	Emitter	0.5v to 0.8v	OL
Base	Collector	0.5v to 0.8v	OL
Emitter	Base	OL	0.5v to 0.8v
Collector	Base	OL	0.5v to 0.8v
Emitter	Collector	OL	OL
Collector	Emitter	OL	OL

Transistor test readings

OL: Over Limit

<u>DIODE TEST USING DIGITAL MULTIMETER</u>

[Diode](#) is a simple PN Junction and two terminal device which allows current to flow though it in one direction (Forward Bias).

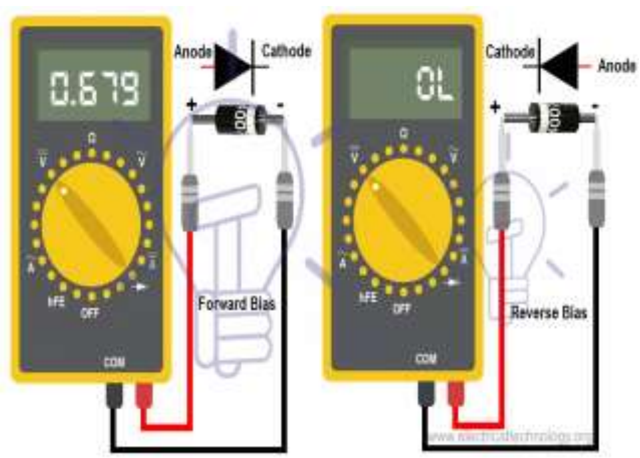


When the cathode terminal of diode is connected to the neutral and anode to positive, It is called in forward bias position and it acts like a short switch where current starts to flow through it. Cathode to positive and anode to neutral is called reverse bias and diode is acts as open switch which is known as reverse bias (This case is reverse in case of zener diode).

Before testing a diode, We must know the diode terminals like Anode (+) and Cathode (-). In most cases, there is a white band color coating on normal PN junctions diodes which indicates as cathode terminal and the rest is anode.

How to Test a Diode using Digital Multimeter?

Testing Diode using DMM (Diode Test Mode + Resistance Mode)



The best practice to test a diode in "Diode test" mode is by measuring the voltage drop across the diode in case of forward biased. Keep in mind that diode in forward-biased acts as closed switch which let to flow current in it like conductors. In reverse-biased diode, it acts like an open switch and doesn't permit current to flow in it as it acts like a resistor.

Forward Biased: When the Positive (Red) test lead is connected to the anode (+) and negative (Black) test lead is connected to the cathode (-) of diode.

Reverse Biased: If we do the reverse as mention above i.e. RED Test lead to cathode (-) and BLACK test lead to the Anode (+) of the diode.

Steps:

- Remove the diode from the circuit i.e. disconnect the power supply across the diode which has to be tested. Discharge all the capacitor (by shorting the capacitor leads) in the circuit (If any).
- Set the meter on "Diode Test" Mode by turning the rotary switch of multimeter.
- Connect the diode leads to the multimeter test leads and note the reading.
- Now, Connect the diode lead to the multimeter test leads in reverse direction (i.e. Reverse the test leads) and note the measurement.
- If the multimeter shows 0.5V – 0.8V for common silicon diodes and 0.2V- 0.3V in case of germanium diodes in the first attempt, its mean the diode is in good condition (forward-biased).

- If multimeter display “OL” in reverse biased, It is good as well.

- If multimeter does not show measurements I.e. if multimeter display “OL” reading in both direction (Forward-biased and reverse-biased), its mean diode is dead and acting as an open switch which doesn’t allow current to flow in it. In case of shorted diode, there will be zero voltage drop across the diode as current will flow through it and it acts like a short path for current. The diode needs to be changed then.
- If the multimeter displays approximately 0.4V in both directions, its mean the diode is short and need to be replaced with new one.